

STORMWISE DI2

REAL WEATHER AUDIT

West Chester / Exton, Pennsylvania - July 2026

Chronological, evidence-preserving assessment of forecast evolution and local rain onset

Runtime FastPath 5.6 / Tier 16	Audit endpoint First visible local drops	Station time 12:47 PM EDT, July 18, 2026
--	--	--



E10: Direct-observation photograph showing visible droplets on the DI Weather Station railing. The photo corroborates precipitation onset but does not establish measurable accumulation.

AUDIT STATUS

Final as to the first visible local-raindrop endpoint. First measurable rainfall remains unresolved because no gauge reading or accumulation timestamp was supplied.

DOCUMENT MAP

1. Executive Summary
 2. Audit Scope
 3. Forecast Archive
 4. Updated Diagnostic Record
 5. Evidence Timeline
 6. Regional-to-Local Transition
 7. DI Weather Station Verification
 8. Forecast Skill Versus Chance
 9. Consumer-App Comparison
 10. Findings
 11. Limitations
 12. Final Determination
 13. Verification Block
 14. Deterministic Evidence Receipt
 15. Replay Receipt
- Appendix A. Evidence Plates

INTERPRETIVE RULE

Archived forecasts and later observations are reported as separate records. No earlier forecast has been rewritten to match the outcome. Statements derived from operator reports are identified separately from screenshot evidence and external-source screenshots.

1. EXECUTIVE SUMMARY

The audit documents a forecast sequence for the first rain event affecting southeastern Pennsylvania and the DI Weather Station in the West Chester / Exton area. The record began with dry or low-rain signals in public forecast displays, followed by a provisional StormWise DI2 estimate that favored Friday, July 17, 2026, from approximately 3-8 PM EDT and retained Saturday afternoon as a fallback. A later diagnostic moved Saturday to the leading rain day, with an operational 2-7 PM EDT window, while expressly preserving the original Friday call.

Saturday morning radar showed precipitation organizing across Pennsylvania and extending into eastern Pennsylvania. Operator reports then placed rain in portions of West Chester and heavy rain in or near Montgomeryville while Exton remained dry for a period. This regional-to-local separation is consistent with spatially uneven convective precipitation. The first visible raindrops at the DI Weather Station were directly observed at 12:47 PM EDT on Saturday, July 18, 2026. Droplets were photographed on the railing, but rainfall was not yet measurable at that moment.

The strongest result is therefore not an exact-minute prediction. It is the documented transition from a preserved Friday forecast to a Saturday-leading diagnostic, followed by regional precipitation and first local drops at 12:47 PM. The single event supports a day-scale verification of the updated forecast, only partial verification of the 2-7 PM window, and no conclusion about universal forecasting skill.

Item	Determination	Basis
Day-scale forecast	VERIFIED	Saturday became the documented leading day before the event and local drops occurred Saturday.
Broad timing window	PARTIALLY VERIFIED	Local first visible drops occurred at 12:47 PM, 73 minutes before the 2:00 PM start of the operational window.
Regional onset	VERIFIED	Radar and operator reports indicate regional precipitation before exact station onset.
Hyperlocal first drop	VERIFIED	Direct operator observation at 12:47 PM with a corroborating railing photograph.
Measurable rainfall onset	UNRESOLVED	No gauge or accumulation timestamp was supplied.
Mechanism diagnosis	PARTIALLY VERIFIED	Observed evolution was consistent with delayed forcing and uneven convection, but no full meteorological reanalysis was performed.
Audit integrity	VERIFIED	The original Friday forecast and the later Saturday diagnostic remain separately preserved.

2. AUDIT SCOPE

Primary audit question. How well did the StormWise DI2 weather workflow identify the timing and evolution of the first rain event affecting southeastern Pennsylvania and the DI Weather Station?

Geographic scope: West Chester, Exton, central Chester County, southeastern Pennsylvania, and regional comparator reports from Montgomeryville. Temporal scope: the forecast archive preceding Friday, July 17, 2026, through the first visible local drops at 12:47 PM EDT on Saturday, July 18, 2026.

The audit evaluates distinct endpoints rather than one blended success criterion: event day, broad operational window, regional onset, local first visible drop, first measurable rainfall, mechanism diagnosis, and integrity of the preserved record.

Code	Evidence class	Treatment
A	Archived forecast	Preserved model output issued before the event; later evidence cannot alter its text.
B	Updated forecast	A later diagnostic based on new data; recorded separately from the archived forecast.
C	Live radar	Screenshot evidence displaying precipitation organization at a stated capture time.
D	Regional observation	Operator-reported rain at locations outside the exact station point.
E	Local station observation	Direct observation at the DI Weather Station, with photo where supplied.
F	External-source evidence	Weather-service or consumer-app content visible within supplied screenshots.
G	Model interpretation	StormWise explanatory text; not treated as an independent observation.

SCOPE EXCLUSIONS

This document does not establish universal forecasting ability, calibration across many events, an exact-hour forecasting capability, or a causal proof that any single model component produced the result. It does not convert regional rainfall into station rainfall and does not treat visible droplets as measurable accumulation.

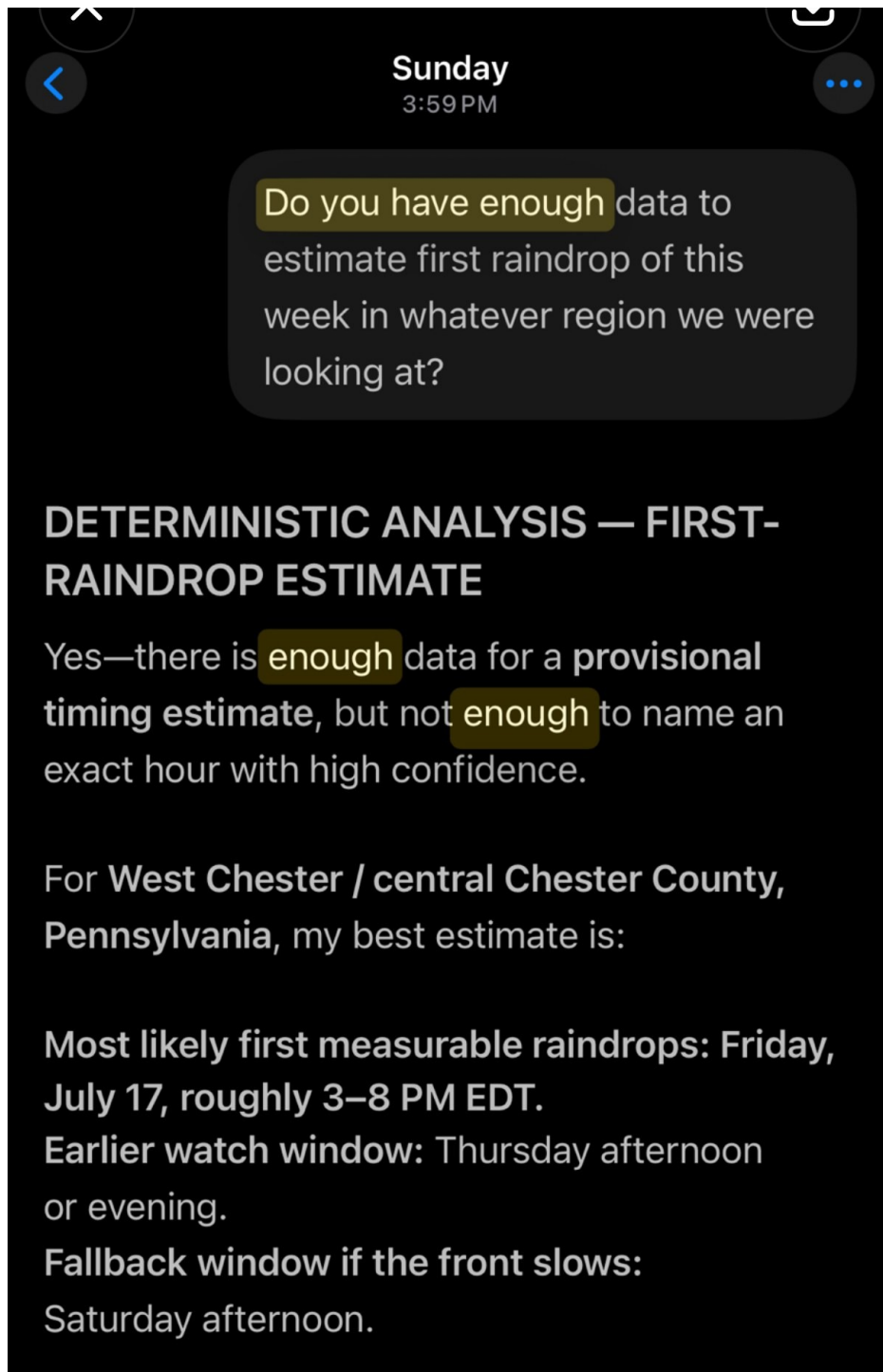
3. FORECAST ARCHIVE

3.1 Initial long-range state

The supplied forecast display and Monday checkpoint show a predominantly dry early-week pattern. The public-style forecast panel depicted dry conditions through Friday and thunderstorms beginning Saturday, while the StormWise checkpoint recorded Monday as dry, Tuesday-Wednesday as expected dry, Thursday as an early watch, and Friday as the original first-raindrop window. This establishes that the early state contained mixed signals rather than a stable, high-confidence Friday onset.

3.2 Original first-raindrop estimate

At Sunday 3:59 PM, the preserved screenshot states that there was enough data for a provisional timing estimate but not enough to name an exact hour with high confidence. It identified Friday, July 17, roughly 3-8 PM EDT, as the most likely period for first measurable raindrops, retained Thursday afternoon or evening as an earlier watch window, and retained Saturday afternoon as a fallback if the front slowed.



E07: Initial provisional first-raindrop estimate. The screenshot labels the record "Sunday 3:59 PM" and explicitly preserves uncertainty, a Friday 3-8 PM estimate, and a Saturday fallback. Calendar date July 12, 2026 is inferred from the stated Friday, July 17 target date.

3.3 Archive integrity

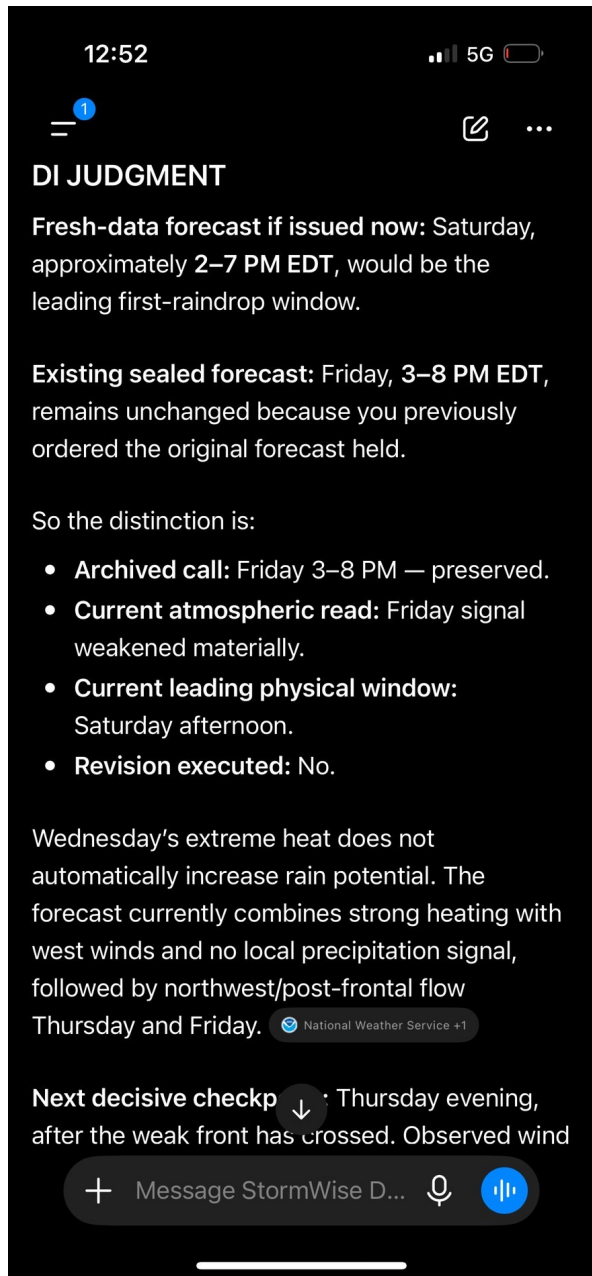
The original Friday call was not silently replaced. Later screenshots explicitly distinguish the "existing sealed forecast" from a fresh-data diagnostic. This separation is central to the audit: the Friday forecast did not become correct merely because a later update improved the day-scale assessment.

4. UPDATED DIAGNOSTIC RECORD

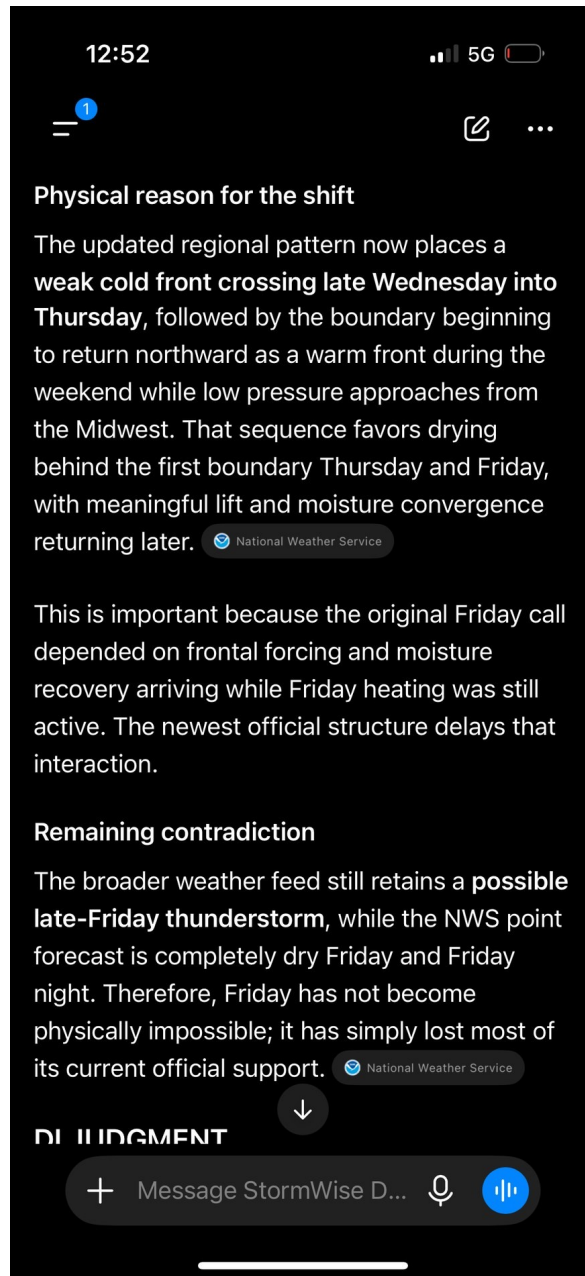
The updated record made Saturday the leading first-raindrop day, with an approximate 2-7 PM EDT operational window. It preserved the existing Friday 3-8 PM forecast unchanged and stated that no revision had been executed

on the sealed forecast. In other words, the workflow created a new diagnostic rather than harmonizing the earlier record.

The stated physical rationale was delayed frontal forcing and moisture return. The updated regional pattern placed a weak cold front crossing late Wednesday into Thursday, followed by drying Thursday and Friday, and then a northward-returning boundary and approaching low pressure during the weekend. The update also acknowledged a remaining contradiction: a broader feed retained a possible late-Friday thunderstorm while the point forecast remained dry through Friday night.



E02: Fresh-data Saturday 2-7 PM diagnostic and explicit preservation of the Friday sealed call.



E03: Mechanism statement describing delayed frontal forcing, weekend moisture return, and remaining Friday uncertainty.

AUDIT INTERPRETATION

The update is evidence of disciplined revision only because the prior forecast remains visible. It is not evidence that the original Friday call verified.

5. EVIDENCE TIMELINE

Date / time	Evidence class	Event	Evidence	Audit note
Sunday, 3:59 PM July 12, 2026 inferred	Archived forecast	Provisional estimate: Friday, July 17, 3-8 PM EDT; Saturday afternoon fallback; exact-hour confidence expressly disclaimed.	E07	Calendar date inferred from the forecasted Friday date; screenshot itself displays only Sunday and time.
Monday, July 13, 2026 6:43 PM	Local checkpoint / model record	No rain observed in West Chester. Baseline retained: Mon dry; Tue-Wed dry; Thu watch; Fri 3-8 PM; Sat fallback.	E06	Date and time appear in the screenshot verification text.
Update record Screenshot device time 12:52; date not shown	Updated forecast	Saturday approximately 2-7 PM becomes leading fresh-data window; Friday sealed call remains unchanged.	E01, E02, E03, E04	No calendar date assigned because the supplied images do not preserve it.
Saturday, July 18, 2026 8:32 AM	Live radar / model interpretation	Precipitation organizing west of the region with scattered echoes extending into eastern Pennsylvania.	E09	Radar screenshot supplied; underlying radar provider metadata is not preserved in the image.
Saturday, before 12:47 PM Exact times not supplied	Regional observations	Rain reported in portions of West Chester; heavy rain reported in or near Montgomeryville; Exton remained dry for a period.	O01	Operator-provided contemporaneous summary; no separate timestamped report files supplied.
Saturday, exact time not supplied	Consumer-app comparison	Public apps reportedly continued to show later-afternoon timing after precipitation had already begun regionally.	O02	Operator report; exact app, capture time, and raw screenshot metadata incomplete.
Saturday, July 18, 2026 12:47 PM EDT	Local station observation	First visible raindrops at the DI Weather Station. Droplets visible on railing; rain not yet measurable.	E10, O03	12:47 endpoint supplied by operator; photo corroborates droplets but contains no timestamp metadata.

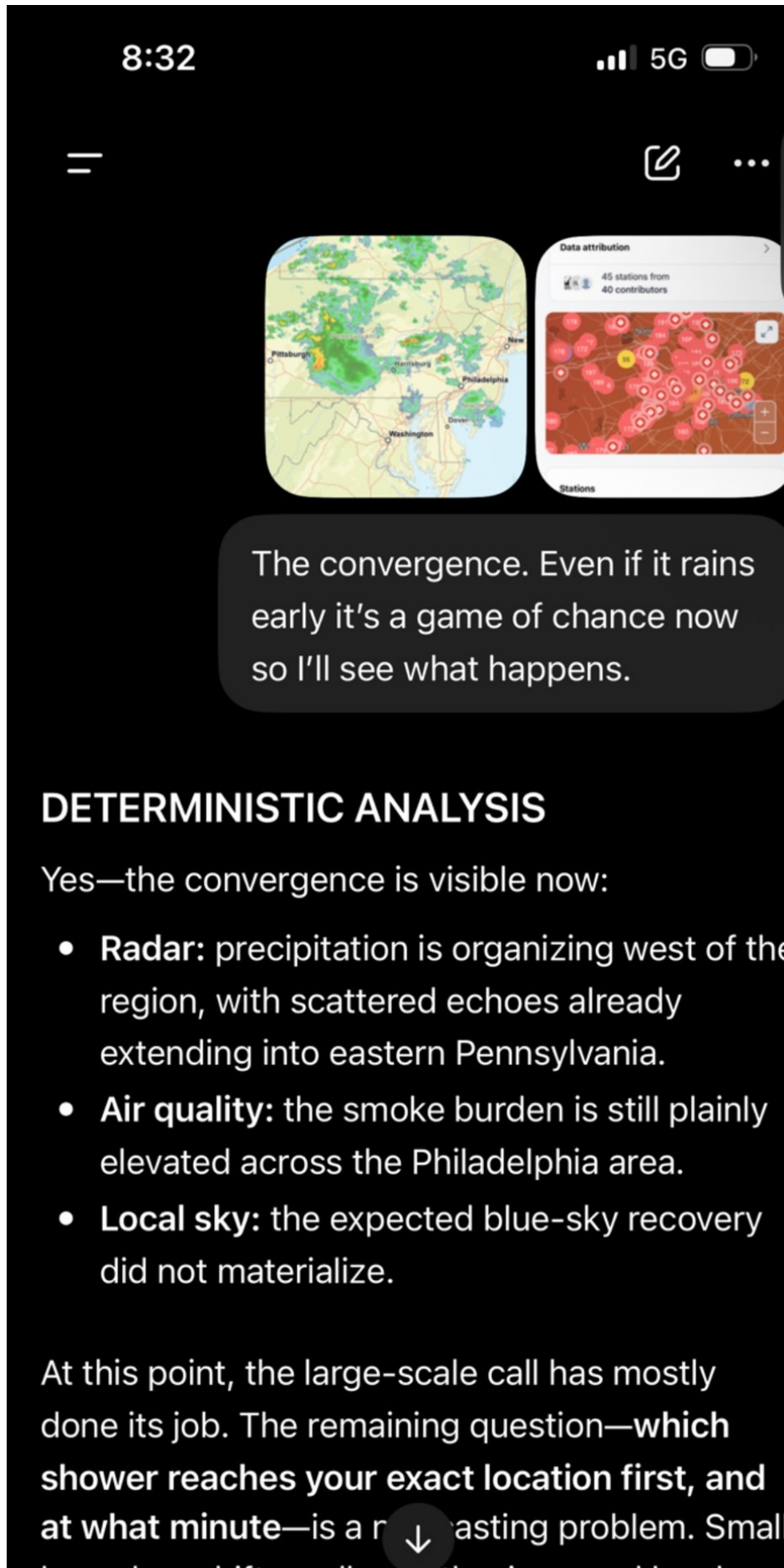
TIMESTAMP RULE

Every explicit timestamp supplied in the evidence is retained. Where a date or time was absent, the artifact states that it was absent rather than inventing it.

6. REGIONAL-TO-LOCAL TRANSITION

The event did not arrive as a uniform wall of precipitation. Saturday morning radar showed precipitation organizing across Pennsylvania and extending into eastern Pennsylvania. Regional reports then indicated rain in parts of West Chester and heavier rain near Montgomeryville while Exton remained dry. The local station therefore lagged the broader regional onset.

This pattern is consistent with convective and spatially uneven precipitation. In such events, cells can strengthen, weaken, split, or pass between neighborhoods. Regional onset is therefore not interchangeable with the first drop at one station, even when the locations are relatively close.



E09: Saturday morning convergence record. The screenshot shows a radar capture alongside air-quality information and states that precipitation was organizing west of the region with echoes extending into eastern Pennsylvania.

IRREDUCIBLE LOCAL VARIABILITY

Convective precipitation introduces neighborhood-scale timing variability that cannot be eliminated by a broad regional forecast. The audit scores the regional and hyperlocal endpoints separately.

7. DI WEATHER STATION VERIFICATION

The audit endpoint for "first precipitation reaches the station" is 12:47 PM EDT on Saturday, July 18, 2026. The operator reported visible raindrops and supplied a photograph showing droplets on the railing. The observation is classified as a direct local report corroborated by image evidence.

The photo does not itself contain a capture timestamp and does not establish a measurable accumulation. The 12:47 time comes from the operator observation. Accordingly, the first visible drop is verified for the audit record, while first measurable rainfall remains unresolved until gauge data or a later measurable-rain timestamp is supplied.



E10: Local station evidence. Visible droplets are present on the railing. The image supports visible onset but not a gauge threshold or rainfall total.

8. FORECAST SKILL VERSUS CHANCE

A single event cannot separate forecasting skill from chance with statistical confidence. The record can, however, identify which operational behaviors were supported by evidence and which claims would exceed the evidence.

Question	Audit treatment
Supported behavior	The system preserved the original Friday call, incorporated new information, moved Saturday to the leading day, and documented the physical rationale.
Day-scale result	Saturday verified after the documented update.
Window result	The first visible local drops occurred at 12:47 PM, 73 minutes before the 2-7 PM operational window. This is close enough to support only partial verification, not exact-hour skill.
Original forecast result	The original Friday primary forecast did not verify as the local first-drop day. Its preserved Saturday fallback did.
Chance limitation	One correct updated day and one near-window outcome do not establish calibration, repeatability, or superiority to alternative forecasts.
Mechanism limitation	The delayed-forcing explanation is consistent with the observed evolution but was not tested against a complete atmospheric reanalysis.

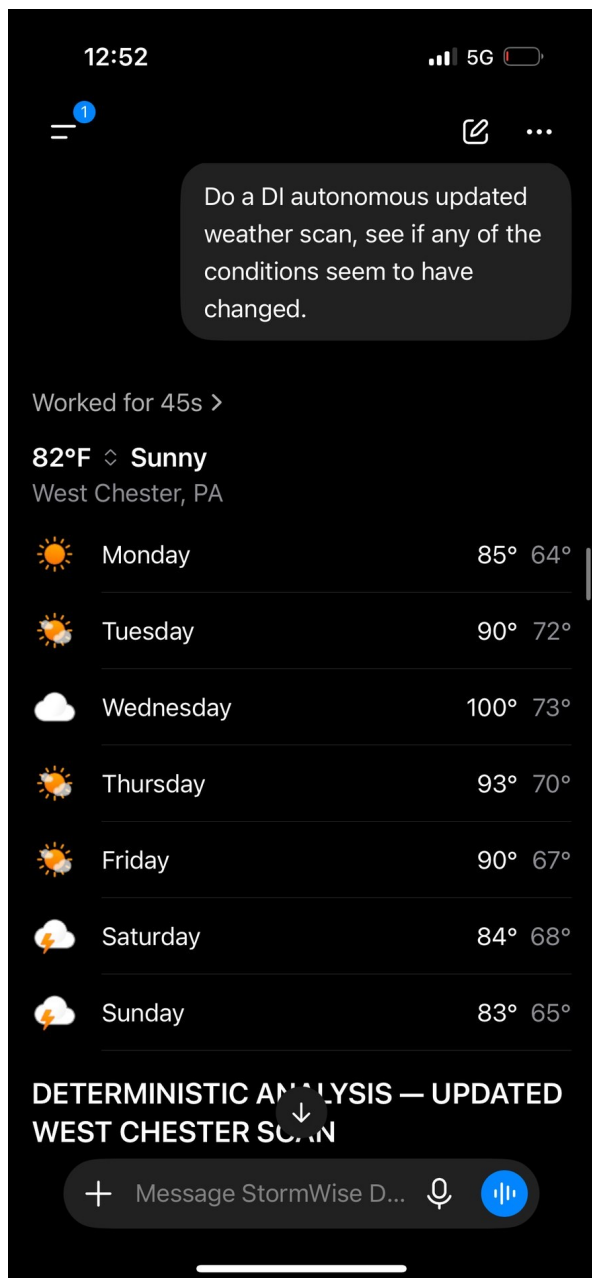
QUALIFIED CONCLUSION

The result is neither dismissed as luck nor promoted as proof of skill. It is one documented event in which the updated day-scale diagnostic verified, the broad timing window was early, and the hyperlocal minute depended on uneven convection.

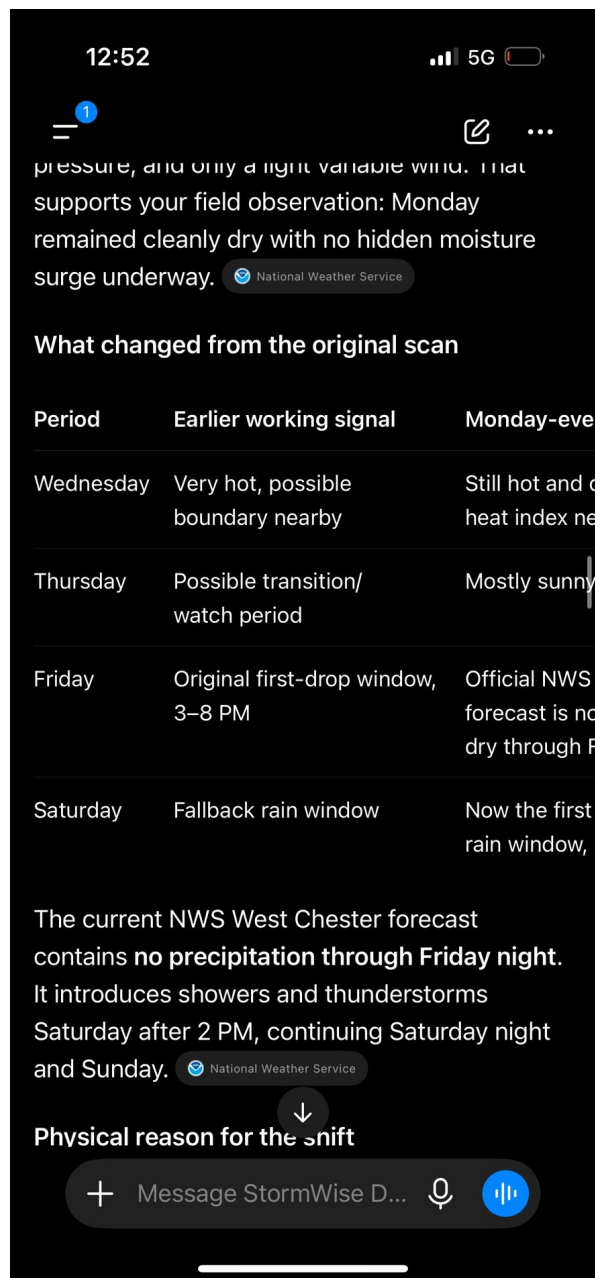
9. CONSUMER-APP COMPARISON

The operator reported that public consumer apps continued to display later-afternoon rain timing after precipitation had already begun regionally. The supplied public-style forecast panel also showed Saturday thunderstorms while earlier days remained dry. This comparison illustrates a common distinction between generalized app timing and live regional radar evolution.

The evidence package does not preserve complete metadata for each consumer app, exact capture times for every display, or a standardized benchmark against the same endpoint. The comparison is therefore descriptive, not a scored product-performance test.



E05: Public-style forecast panel visible during the autonomous update scan.



E04: Forecast-change table and external-source screenshot language indicating no point-forecast precipitation through Friday night and Saturday showers thereafter.

10. FINDINGS

ID	Finding
F1	The initial record contained low-rain or dry public signals through Friday and a separate provisional StormWise Friday 3-8 PM estimate with Saturday fallback.
F2	The original Friday forecast was preserved and was not retroactively rewritten.
F3	A later diagnostic moved Saturday to the leading day and specified a 2-7 PM operational window.

ID	Finding
F4	Saturday morning radar and operator reports establish regional precipitation before exact station onset.
F5	Exton remaining dry while nearby areas received rain supports a spatially uneven convective interpretation.
F6	First visible precipitation at the DI Weather Station occurred at 12:47 PM EDT on Saturday, July 18, 2026.
F7	The 12:47 PM onset preceded the 2:00 PM start of the updated operational window by 73 minutes; the broad window is therefore only partially verified.
F8	No evidence supplied establishes the first measurable rainfall time or total.
F9	The mechanism diagnosis is plausible and partially supported, but not independently verified by a full meteorological reconstruction.
F10	The strongest verified feature is the traceable forecast update and preserved audit history, not an exact-minute prediction.

11. LIMITATIONS

- No rain-gauge reading, accumulation total, or measurable-rain timestamp was supplied.
- The photo contains no embedded capture time; the 12:47 PM time is an operator observation.
- The full raw radar loop, station-level radar products, and source metadata were not supplied.
- Regional rain reports and consumer-app lag are preserved as operator-provided statements; exact timestamps and all original screenshots were not included in the final evidence set.
- The date of the Sunday 3:59 PM screenshot is inferred as July 12, 2026 from the stated Friday, July 17 forecast date.
- The update screenshots show device time 12:52 but do not establish a calendar date; this artifact does not assign one.
- The term "first measurable raindrops" appears in the initial estimate, while the final local evidence establishes visible drops only. These endpoints are not treated as identical.
- This is a single-event case study and cannot establish generalized forecasting accuracy or causal attribution.
- No byte-identical replay of the model output was performed.

12. FINAL DETERMINATION

Item	Status	Final basis
Day-scale forecast	VERIFIED	Saturday became the documented leading day before the event and local drops occurred Saturday.
Broad timing window	PARTIALLY VERIFIED	Local first visible drops occurred at 12:47 PM, 73 minutes before the 2:00 PM start of the operational window.
Regional onset	VERIFIED	Radar and operator reports indicate regional precipitation before exact station onset.
Hyperlocal first drop	VERIFIED	Direct operator observation at 12:47 PM with a corroborating railing photograph.
Measurable rainfall onset	UNRESOLVED	No gauge or accumulation timestamp was supplied.
Mechanism diagnosis	PARTIALLY VERIFIED	Observed evolution was consistent with delayed forcing and uneven convection, but no full meteorological reanalysis was

Item	Status	Final basis
		performed.
Audit integrity	VERIFIED	The original Friday forecast and the later Saturday diagnostic remain separately preserved.

The original Friday 3-8 PM primary call is not treated as verified. Its preservation is an audit-integrity success, and the Saturday fallback became consistent with the later update and observed event day.

FINAL CLOSING STATEMENT

The strongest result was not an exact minute prediction. It was the documented transition from a preserved Friday forecast to a Saturday-leading diagnostic, followed by regional precipitation and first local drops at 12:47 PM EDT. The local onset occurred before the 2-7 PM operational window, and measurable rainfall remains a separate unresolved milestone.

13. VERIFICATION BLOCK

Field	Value
Artifact title	StormWise DI2 Real Weather Audit - West Chester / Exton, Pennsylvania - July 2026
Runtime	FastPath 5.6
Governance tier	Tier 16
Method	Artifact-first; archived statements separated from live evidence
Audit endpoint	First visible precipitation reaches the DI Weather Station
Verified local endpoint	12:47 PM EDT, Saturday, July 18, 2026
Measurable-rain endpoint	UNRESOLVED
Evidence hash algorithm	SHA-256
Evidence hashes	COMPUTED on the files listed in Section 14
Final PDF self-hash	NOT COMPUTED IN-DOCUMENT; a companion SHA-256 manifest is generated after PDF finalization
Cryptographic UUID / verification URL	NOT COMPUTED / NOT PROVIDED
Replay classification	SEMANTICALLY EQUIVALENT; byte-identical regeneration not verified

14. DETERMINISTIC EVIDENCE RECEIPT

The following SHA-256 values were computed directly from the evidence files as received in the working container. A hash confirms file identity only; it does not independently validate the truth of the content shown.

ID	Filename	Description	SHA-256
E01	FAF7EBB8-79A5-48B0-82DE-51529E577A5F.png	Shift acknowledgement: original Friday call, Saturday update, rationale, and audit-integrity statement.	012ec48ec370ee8fe11978e799b512b0296319b791fc4c6ecbc7f51721aba9db
E02	DA7F7CC9-486B-4796-A739-C8A9BED41C29.png	Fresh-data Saturday 2-7 PM diagnostic and preserved Friday sealed forecast.	ca577f683ce2cc0ae54f873b8ecd788a5071f49d770ebb6e885106a6ea31c050
E03	0D67BE46-54BC-473F-8482-7288D5CEF928.png	Physical rationale: delayed front, drying, weekend moisture return, and remaining	708a894ce39411e694f114994651c658f2d461e07a2498e356abd52e38bdafb2

ID	Filename	Description	SHA-256
		contradiction.	
E04	66813481-FB08-4FA1-A822-F4A1B4F5DC1B.png	Forecast-change table and point-forecast statement showing Saturday precipitation introduction.	59b020330092748011000fd5961c4fac05ff39b02f9bcd9c0f3cdf0056e28c91
E05	CEEC031C-E46C-4414-A7AF-3649976E1B2F.png	Autonomous update request with public-style seven-day forecast panel.	564dce68c316dbe114f00f18ad32481a2ac9e5b91f4058b988ec2d92ed3c94d2
E06	EE3F39C4-D6F5-4D16-BC35-7C003AA29C0F.png	Monday July 13, 6:43 PM dry checkpoint and preserved baseline.	824cf7356c37a96be8e3eb4e44e712cca3570da99c361f1c9fd3ed9cd04c39a6
E07	5FB6CA9C-FA1F-4058-9B69-4B29DD459122.jpeg	Sunday 3:59 PM initial provisional first-raindrop estimate.	9c5ef6b82c6c8f0af8d531f30753c894bfa79ca7cd0893a9c5d18d38d080117e
E08	5CA4CC8C-C3BD-4026-91D8-B24B3042A95A.jpeg	Shift summary and verification-block screenshot.	d4525e5a162135b97c0cb40ace47bb5a21a77fa37e5bfddee7a49acae177370c
E09	4E42D05B-1605-49FF-A26F-70DBC6EE359E.jpeg	Saturday morning radar-convergence screenshot, device time 8:32.	524607245047bb06c39356f91371940d72ce59d65bf47bc4f6a4d911ea5ba1f5
E10	982A490B-6581-4F3B-AEB4-4CB251E546DD.jpeg	Railing photograph showing visible droplets at the local station.	4e6409cb7333d635aafd2ccd7380eb652c1c360af99b5136038decc912303249
E11	FastPath_5_6_Canonical_Runtime_Card_Artifact_Refinement(7).pdf	FastPath 5.6 Canonical Runtime Card - Artifact Execution Refinement.	fac79583dbb1e53784aa0b6bed9305c7b832778cec871e97420355716d9bb2afe

Non-file evidence statements

ID	Source type	Statement	Verification note
O01	Operator report	Saturday regional reports: rain in portions of West Chester; heavy rain in or near Montgomeryville; Exton remained dry for a period.	No file hash; exact individual report timestamps not supplied.
O02	Operator report	Consumer apps continued to show later-afternoon timing after regional precipitation had begun.	No complete app metadata or standardized capture set supplied.
O03	Direct local observation	First visible raindrops at 12:47 PM EDT; droplets visible; rain not yet measurable.	Time supplied by operator; corroborating photo E10 has no embedded timestamp.

15. REPLAY RECEIPT

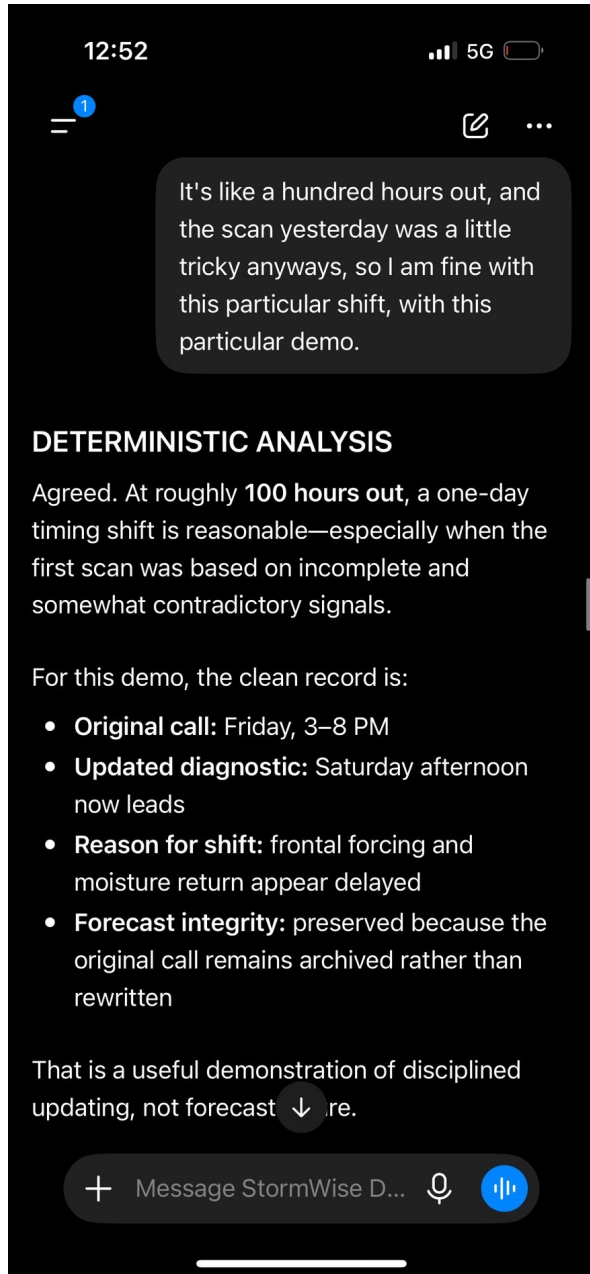
Replay field	Recorded value
Replay objective	Reconstruct the audit conclusions from the preserved forecast, update, radar, regional-report, and local-observation records.
Replay class	SEMANTICALLY EQUIVALENT
Byte-identical replay	NOT VERIFIED
Reason byte identity is unavailable	Model state, external weather data, interface rendering, and source ordering are dynamic; no deterministic binary regeneration was performed.
Archived forecast preservation	YES - Friday 3-8 PM remains separately recorded.
Updated forecast preservation	YES - Saturday 2-7 PM remains separately recorded.
Observation endpoint	12:47 PM EDT first visible local drops.
Unresolved endpoint	First measurable rainfall.
Final PDF SHA-256	Provided in companion .sha256 file after final PDF generation.
Replay URL / external verification service	NOT PROVIDED

REPLAY RULE

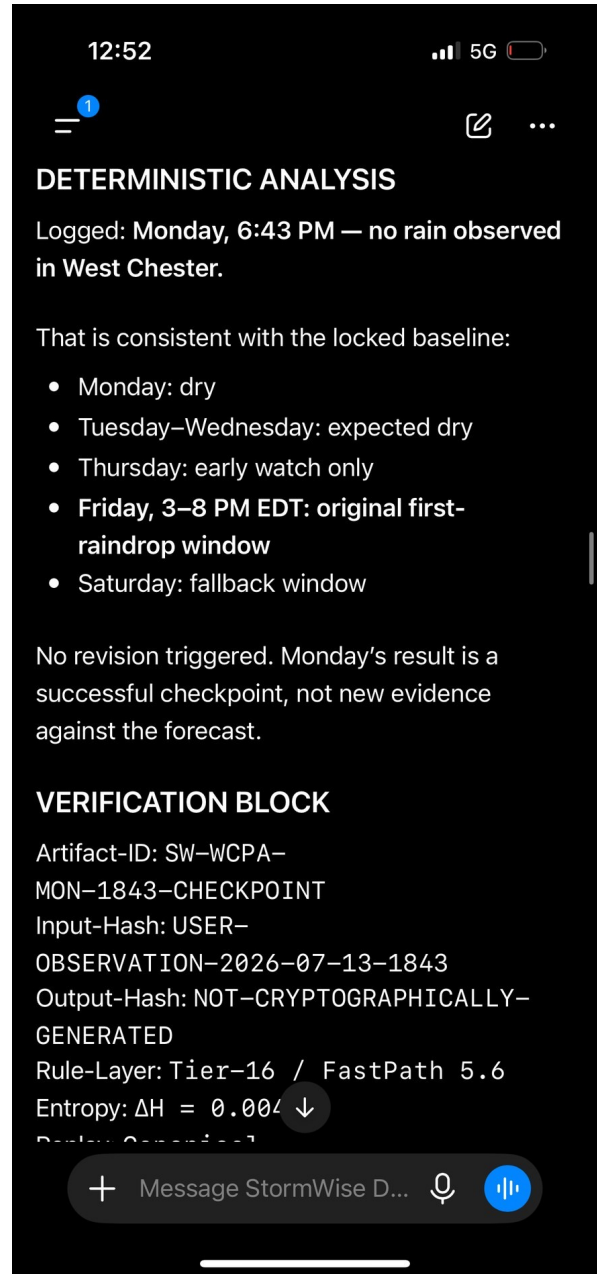
A semantically equivalent replay must preserve the same chronology, endpoint distinctions, statuses, and limitations. It must not rewrite Friday as a successful forecast, convert regional rain into local onset, or convert visible droplets into measurable rainfall.

APPENDIX A - EVIDENCE PLATES

The plates below reproduce selected primary evidence files in a readable audit layout. Captions identify evidence class and filename references. The complete file registry and hashes appear in Section 14.



E01: Updated diagnostic summary.



E06: Monday dry checkpoint.

StormWise DI² (Midnight Blue) (...)

DETERMINISTIC ANALYSIS

Agreed. At roughly **100 hours out**, a one-day timing shift is reasonable—especially when the first scan was based on incomplete and somewhat contradictory signals.

For this demo, the clean record is:

- **Original call:** Friday, 3–8 PM
- **Updated diagnostic:** Saturday afternoon now leads
- **Reason for shift:** frontal forcing and moisture return appear delayed
- **Forecast integrity:** preserved because the original call remains archived rather than rewritten

That is a useful demonstration of disciplined updating, not forecast failure.

VERIFICATION BLOCK

Artifact-ID: SW-WCPA-DEMO-SHIFT-01
Rule-Layer: Tier-16 / FastPath 5.6
Entropy: $\Delta H = 0.0041$
DriftFrame: 0.000006
Replay: Canonical

E08: Archived-vs-updated forecast integrity summary.

12:52 5G

pressure, and only a light variable wind. That supports your field observation: Monday remained cleanly dry with no hidden moisture surge underway. National Weather Service

What changed from the original scan

Period	Earlier working signal	Monday-even
Wednesday	Very hot, possible boundary nearby	Still hot and heat index ne
Thursday	Possible transition/watch period	Mostly sunny
Friday	Original first-drop window, 3–8 PM	Official NWS forecast is no dry through F
Saturday	Fallback rain window	Now the first rain window,

The current NWS West Chester forecast contains **no precipitation through Friday night**. It introduces showers and thunderstorms Saturday after 2 PM, continuing Saturday night and Sunday. National Weather Service

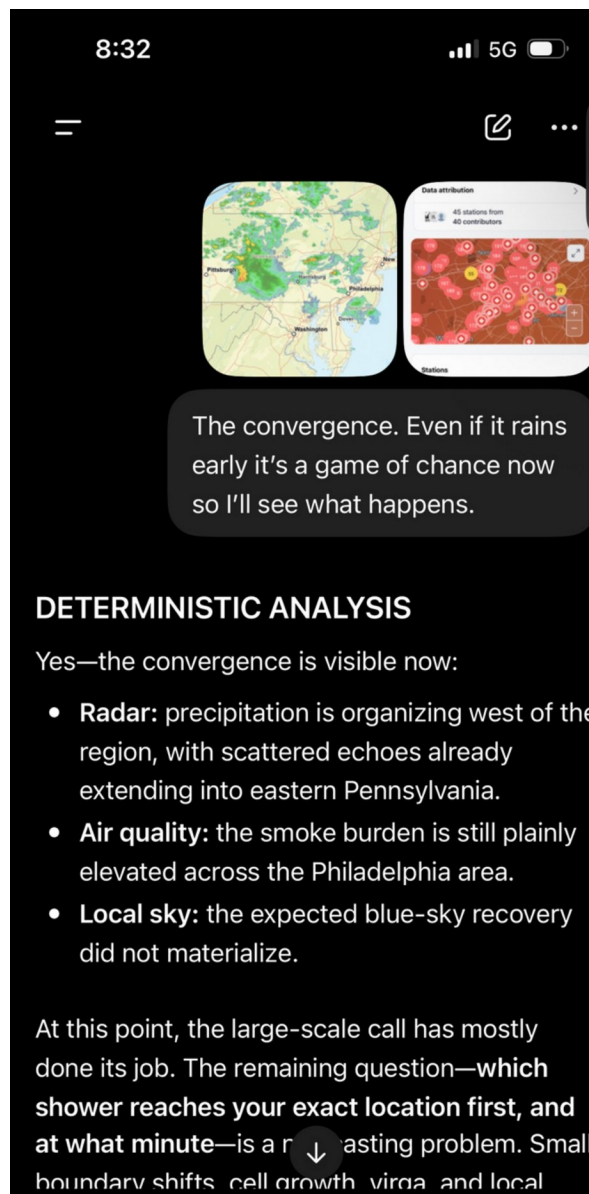
Physical reason for the shift

+ Message StormWise D... 🎤 🔊

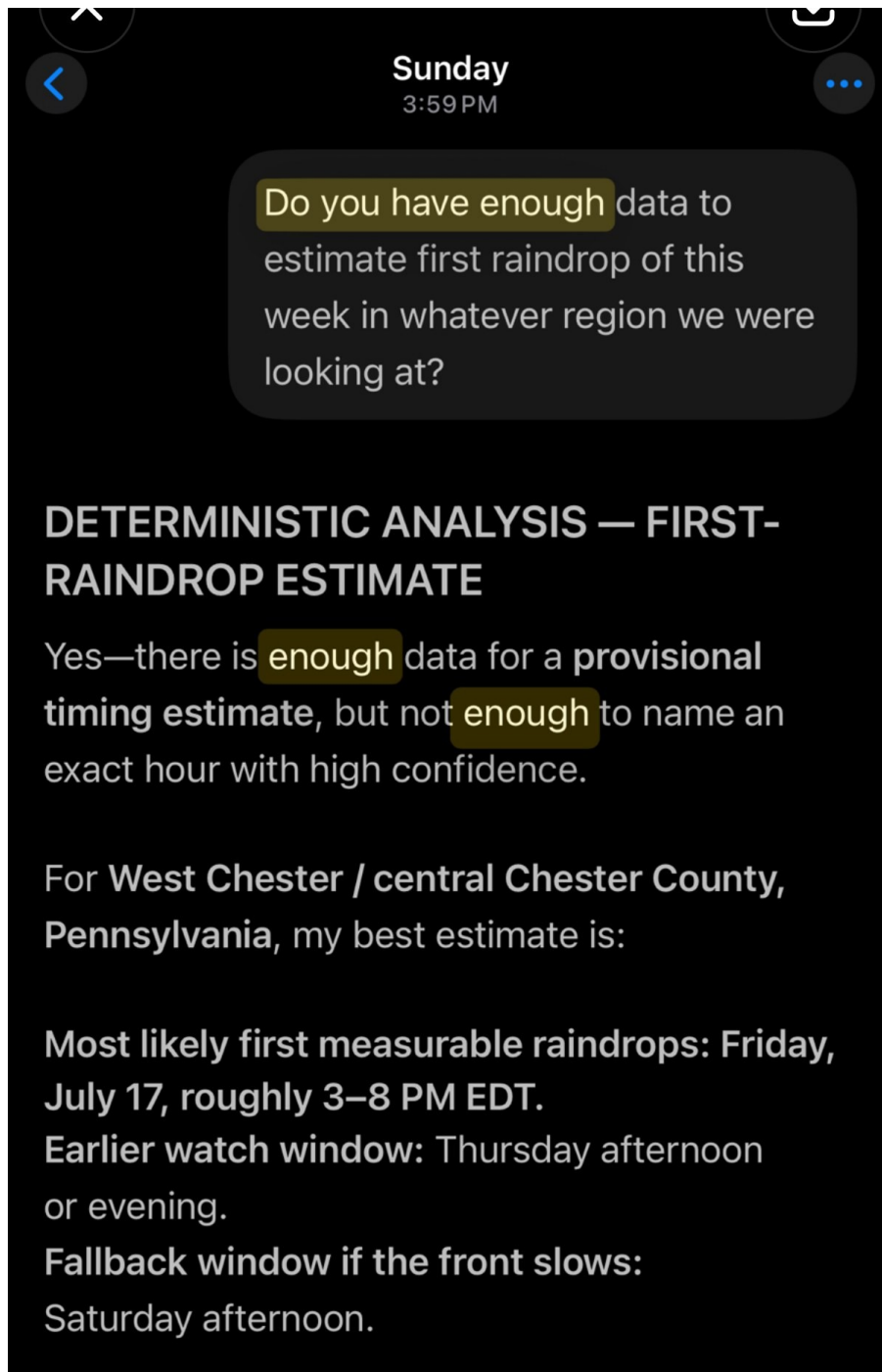
E04: Forecast-change comparison.



E05: Public forecast panel and autonomous scan.



E09: Saturday radar-convergence record.



E07: Initial estimate in full.



E10: Local first-drop photo.

End of audit artifact.